SECTION 1 – PROJECT OVERVIEW

This project involves the analysis of Global Content Insights for Netflix, a streaming service based on a monthly subscription that provides access to a wide range of TV series, movies and documentaries through internet-connected devices worldwide.

The analysis followed the following steps:

1. **Data Collection**
2. **Data Cleaning**
3. **Data Analysis**
4. **Key Findings and KPIs**
5. **Data Visualisation**
6. **Insights**

The aim of this report is to provide an overview of content preferences over the 2016–2025 period, focusing on the relationship between IMDb ratings and views (millions), as well as content distribution by country and genre performance.

SECTION 2 – OBJECTIVES

The objective of this analysis is to identify key patterns and insights within the Netflix content dataset that could help better understand content performance and audience preferences.

The report investigates whether there is a relationship between IMDb ratings and views, specifically whether higher-rated content appears to receive more views. In addition, the analysis examines differences between countries, the performance of different genres, and the distribution of titles across countries. Greater attention is given to countries and genres with a higher number of titles, as larger samples can provide a more meaningful basis for comparison and reduce the risk of overinterpreting results from categories with very few observations.

SECTION 3 – DATASET

The dataset used for this analysis was obtained from Kaggle and contains 200 titles/records distributed across 10 columns. The data covers the period from 2016 to 2025 and includes information such as Year, Title, IMDb rating, Viewership, Directors/Creators, Lead Actors, Release Date, Duration, Country and Genre.

Among these variables, IMDb rating, Viewership, Country and Genre were the main focus of the analysis

- **IMDb ratings** - used as an indicator of audience ratings.
- **Viewership** – analysed to investigate whether there was a relationship between the number of views and IMDb ratings
- **Country and Genre** - examined to identify differences in content distribution and performance across different categories.

The dataset was considered suitable for this project because it combines ratings, viewership and content characteristics, allowing different aspects of content performance to be explored. Particular attention was given to countries and genres with a higher number of titles, as larger samples provide a more representative basis for comparison and reduce the risk of overinterpreting results from categories with very few observations.

SECTION 4 – DATA CLEANING

In this section, the process of transforming the original dataset into a suitable format for analysis is explained.

Firstly, no missing values were identified in the dataset.

Out of the 200 records, 142 contained usable Viewership values and were therefore included in the views-based analysis, while the remaining records were kept in the dataset and were not deleted.

The original Viewership values were presented in a mixed format, containing both numerical and textual information. These values were cleaned and converted into numerical values expressed in millions. A new standardised column, called **Views Million**, was therefore created to make the data suitable for calculations and further analysis.

The formula used was:

=VALUE(TEXTBEFORE(TEXTAFTER(H2,";")," min"))

This procedure made it possible to use the cleaned values for further analysis, including the calculation of average, minimum and maximum views and the correlation between IMDb ratings and views.

In addition, data types and consistency were checked to ensure that the numerical values could be used correctly. Some extreme values were also identified in the dataset, such as a maximum value of **482 million views**.

SECTION 5 – ANALYSIS & METHODOLOGY

The analysis and methodology followed five main steps:

1. **Descriptive statistics**
2. **Correlation analysis**
3. **Country analysis**
4. **Genre analysis**
5. **Visualisation**

1. *Descriptive statistics*

Considering the total number of the titles in the dataset (200), descriptive statistics were used to examine the minimum, maximum and average values for Views and IMDb ratings

For Views, the minimum value was **3.2 million**, while the maximum value was **482 million**, highlighting a considerable difference between the lowest and highest reported values. The average number of views was calculated using the formula:

=AVERAGE(L2:L201)

The same approach was used to calculate the minimum, maximum and average IMDb rating. The average IMDb rating was calculated using:

=AVERAGE(C2:C201)

These measures provided a general overview of the distribution of Views and IMDb ratings before carrying out further analysis.

2. *Correlation analysis*

Correlation analysis was considered a central part of the analysis because it was used to investigate whether there was a linear relationship between Views and IMDb ratings.

The following formula was used:

=CORREL(L2:L201,C2:C201)

where L2:L201 represents the Views Million column and C2:C201 represents the IMDb rating column. The resulting correlation coefficient was **0.0064**.

3. *Country analysis*

To develop the country analysis, PivotTables were created to compare the countries with the highest number of titles. A separate PivotTable was also used to compare the average IMDb ratings across these countries.

The analysis focused on countries with a higher number of titles in order to provide a more meaningful basis for comparison and reduce the risk of overinterpreting results from countries represented by very few observations.

4. *Genre analysis*

A PivotTable was also created to examine genre performance.

In this analysis, **Crime Drama** and **Coming-of-age** were included because they met the selected criterion of having **five or more titles**. This approach was used to focus on genres with a larger number of observations. The analysis included the number of titles, average IMDb rating and average Views Million for each genre.

5. *Visualisation*

The data was represented through a selection of charts and graphs to make the results easier to compare and interpret.

Two **clustered column charts** were used for the country analysis. The first chart compares countries based on the number of titles, while the second compares their average IMDb ratings.

A third visualisation, a **scatter plot**, was used to display the relationship between IMDb ratings and Views Million. The distribution of the data points provides a visual representation of the relationship between the two variables and complements the correlation analysis.

SECTION 6 – KEY FINDINGS

1. *Overall Performance*

The overall analysis of the dataset shows that 142 usable Viewership values were available out of 200 titles. The average number of views was **49.15 million**, based on a minimum of **3.2 million** and a maximum of **482 million** views. The average IMDb rating across the dataset was **7.70**.

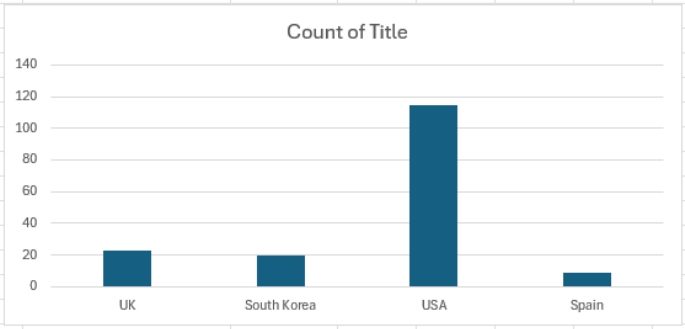
2. *IMDb vs Views*

The scatter plot and correlation analysis show a correlation coefficient of **0.0064**, indicating virtually no linear correlation between IMDb ratings and Views Million within this dataset.

3. *Countries*

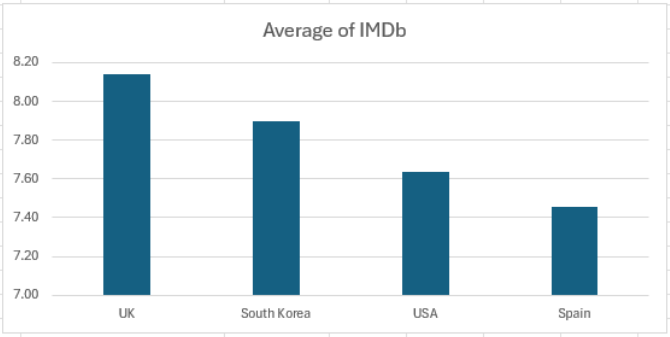
The first clustered column chart highlights the USA as the country with the highest number of titles, with 115 titles:

Country Analysis	Count of Title
UK	23
South Korea	20
USA	115
Spain	9



The second clustered column chart shows that the **UK** has the highest average IMDb rating among the four countries analysed, with an average rating of **8.14**. South Korea follows with **7.90**, while the USA and Spain have average IMDb ratings of **7.64** and **7.46**, respectively:

Country	Average of IMDb
UK	8.14
South Korea	7.90
USA	7.64
Spain	7.46



4. Genres

The PivotTable shows that **Crime Drama** and **Coming-of-age** have the same average IMDb rating of **8.18**, despite being represented by a different number of titles. Crime Drama includes **13 titles**, while Coming-of-age includes **5 titles**.

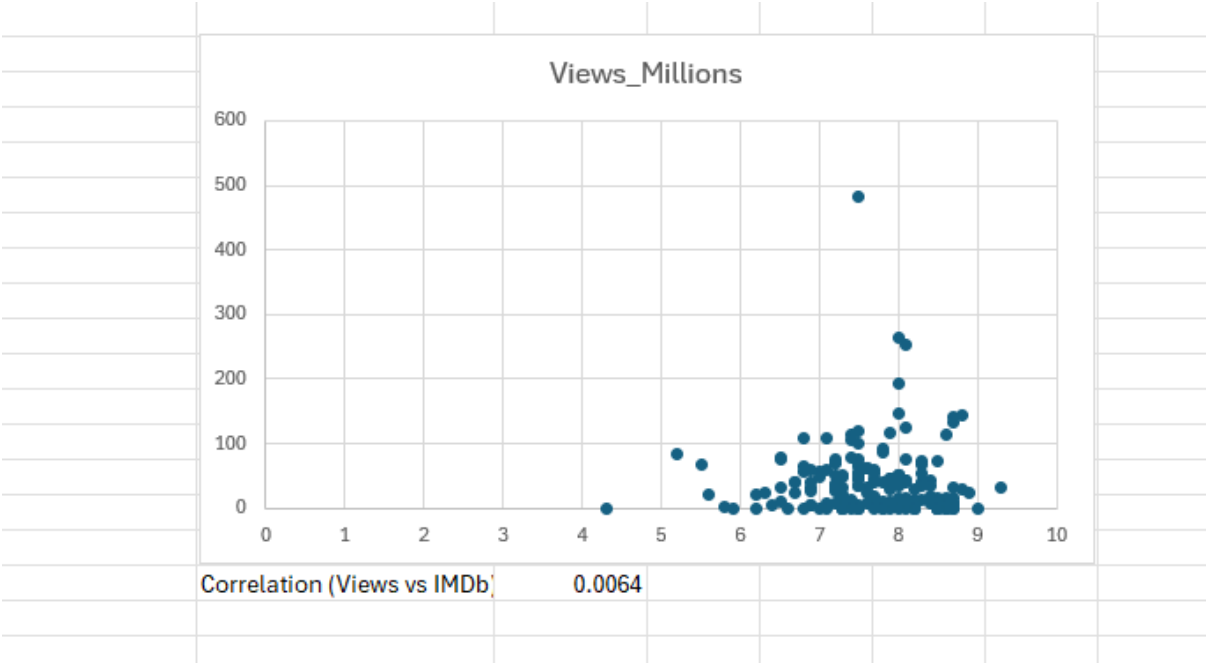
When comparing average Views Million, Crime Drama has the higher average, with **43.06 million views**, compared with **35.62 million** for Coming-of-age:

Genre Performance			
Genres	**Titles**	**Average of IMDb**	**Average of Views_Millions**
Crime Drama	13	8.18	43.06
Coming-of-age	5	8.18	35.62

SECTION 7 – INSIGHTS & INTERPRETATION

1. *IMDb vs Views*

In this dataset, the scatter plot represents the relationship between IMDb ratings and Views Million. The correlation coefficient of **0.0064** indicates virtually no linear association between the two variables. Therefore, higher IMDb ratings do not appear to be associated with higher views within this dataset. This suggests that viewership and IMDb ratings follow different patterns in the analysed data.



2. *Country representation vs rating*

The PivotTables comparing the number of titles and average IMDb ratings show that a higher number of titles does not necessarily correspond to a higher average IMDb rating. The USA has the highest number of titles, with **115**, while the UK has the highest average IMDb rating among the four countries analysed, at **8.14**.

Country Analysis	Count of Title
UK	23
South Korea	20
USA	115
Spain	9

Country	Average of IMDb
UK	8.14
South Korea	7.90
USA	7.64
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The other countries show different patterns: South Korea has **20 titles** with an average IMDb rating of **7.90**, while Spain has **9 titles** with an average IMDb rating of **7.46**. This highlights the difference between the level of representation of a country in the dataset and its average IMDb rating.

3. *Genre Performance*

Crime Drama and Coming-of-age have different numbers of titles, with **13** and **5** titles respectively, but both have the same average IMDb rating of **8.18**. At the same time, their average Views Million differs, with Crime Drama reaching **43.06 million** compared with **35.62 million** for Coming-of-age.

This shows that similar average IMDb ratings do not necessarily correspond to the same level of viewership.

4. *Importance of sample size*

The analysis also highlights the importance of considering the number of observations within each category. A genre represented by only one title can have a very high average number of views simply because that individual title received many views. For this reason, comparisons based on categories with a larger number of observations can provide a more meaningful basis for analysis and reduce the risk of overinterpreting results from categories with very few titles.

SECTION 8 – LIMITATIONS

Although the analysis provides useful insights, there are some limitations that should be considered.

The dataset is not large enough to represent the entire global landscape, as it contains only 200 titles. Furthermore, not all 200 titles had a usable value for analysing Views, meaning that the views-based analysis was conducted using 142 titles.

The results also have limitations when interpreting relationships between variables. The correlation analysis does not allow conclusions about causation. In addition, categories with only a few titles cannot be considered as reliable for comparison as categories with a larger number of observations.

Furthermore, the average IMDb rating of 7.70 does not, on its own, determine a high level of audience satisfaction, as IMDb rating is only one indicator of audience ratings and does not capture all aspects of satisfaction.

There might even be other factors that influence the views and are not considered in the database: Age of the audience and time of the year.

The age of the audience is an important factor as it can show a correlation between age of the population and related views and IMDb rating; adults may be more likely to watch Crime drama

than teenagers, while younger audiences may have different preferences for genres such as Coming-of-age compared with adults.

The time of year could also influence viewership. If data showing the number of views across different seasons were available, it would be possible to establish whether certain seasons are associated with higher levels of viewership than others. For example, during the summer, people may be less likely to watch TV compared with the winter, when colder weather may encourage people to spend more time at home relaxing.

SECTION 9 – CONCLUSIONS

The main aim of this project was to investigate whether there was any correlation between IMDb ratings and Views Million through the analysis of the Global Content Insights dataset. The correlation analysis, based on 142 usable Viewership values out of 200 total titles, resulted in a correlation coefficient of 0.0064, indicating virtually no linear correlation between the two variables.

The analysis of average IMDb ratings and genre performance also highlighted that a sufficient number of observations is important when comparing categories, as categories represented by only a few titles may provide less reliable insights.

From my personal point of view, this project has been a valuable way to improve my analytical skills in the field of data analysis. I developed my ability to use PivotTables, select suitable charts for different types of data, and carry out data cleaning and analysis in Excel. The project also helped me understand the importance of considering the limitations of a dataset when interpreting results.